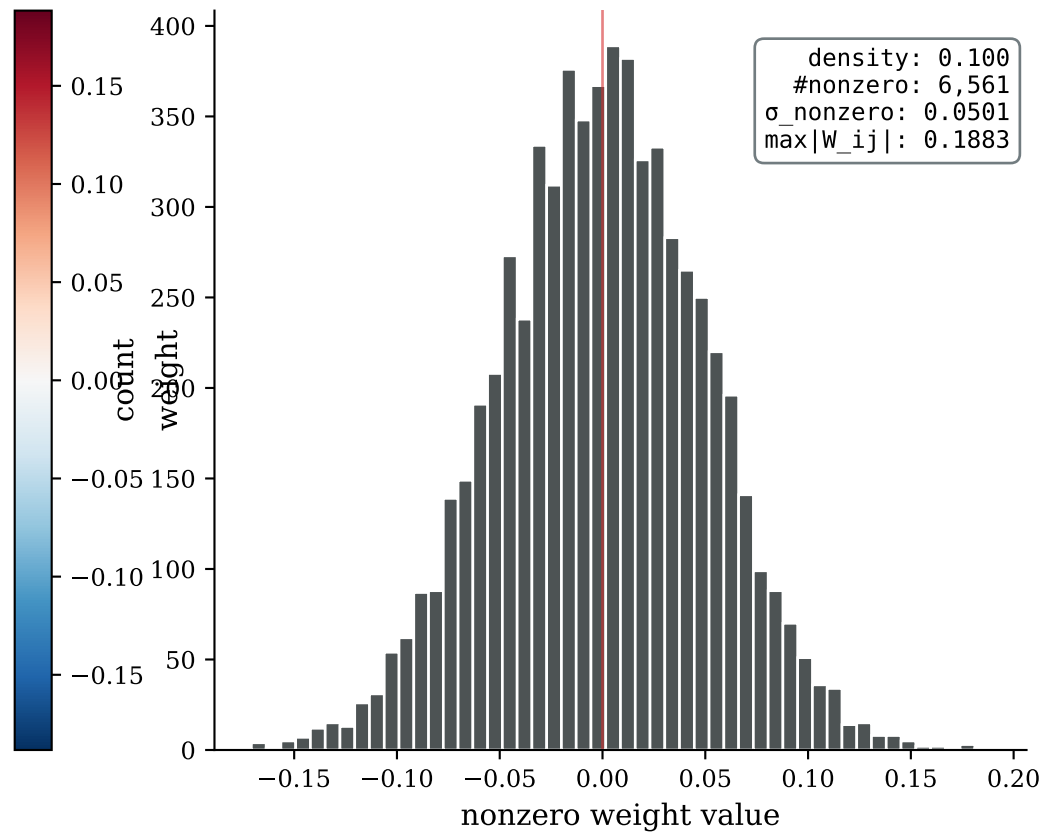
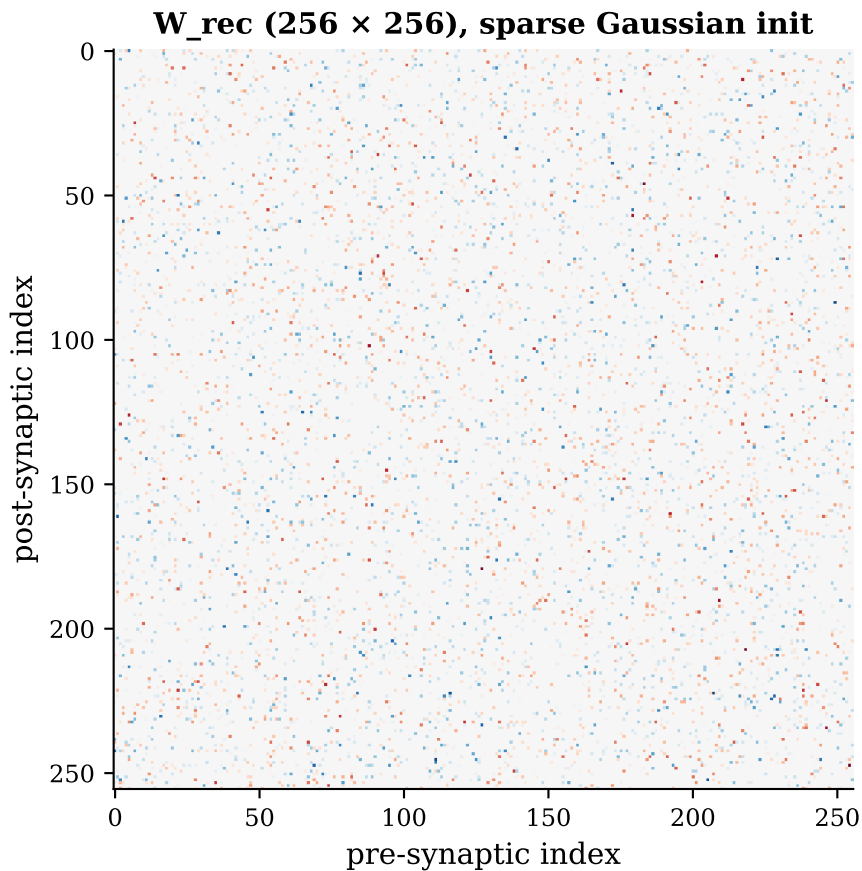
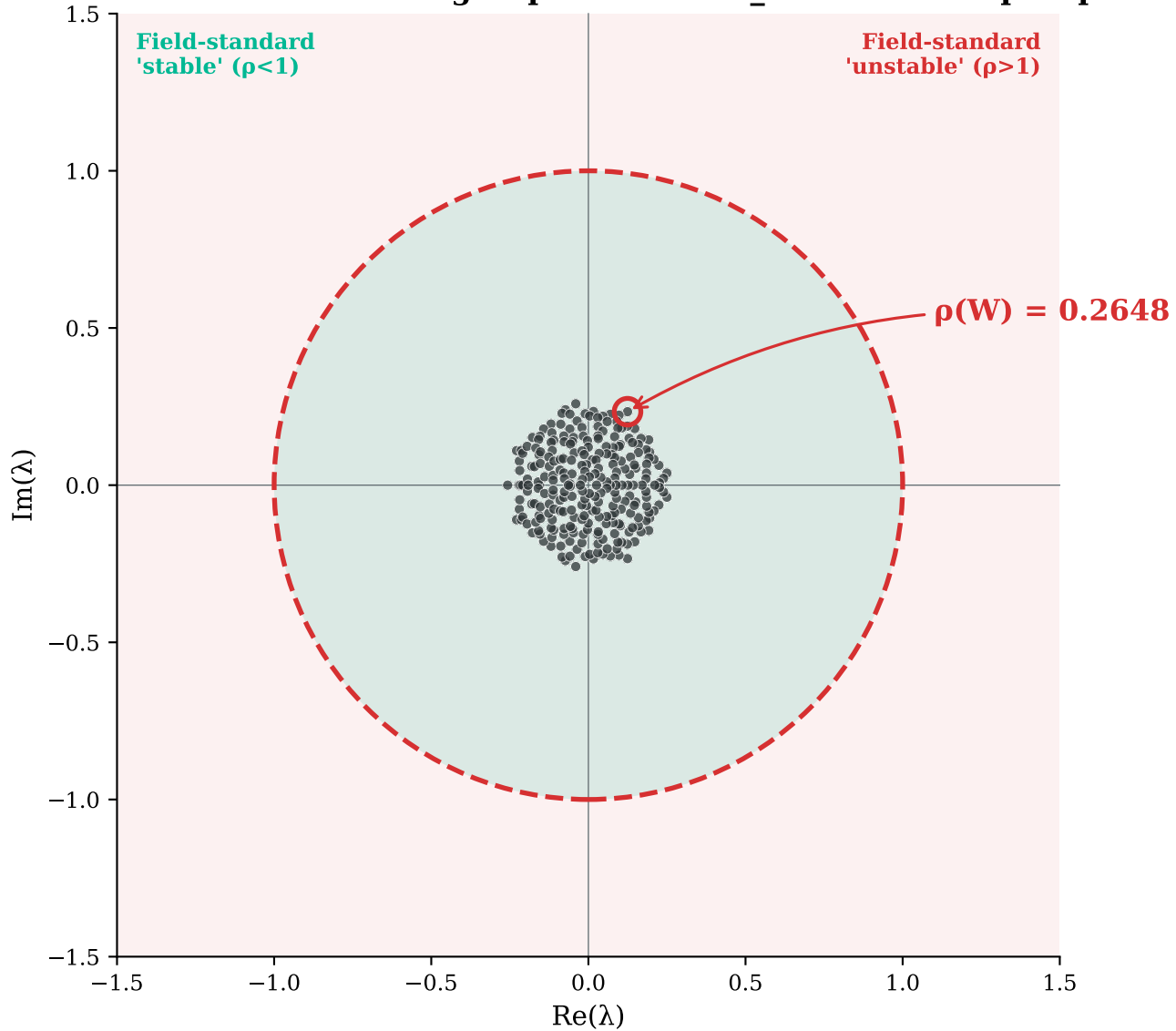


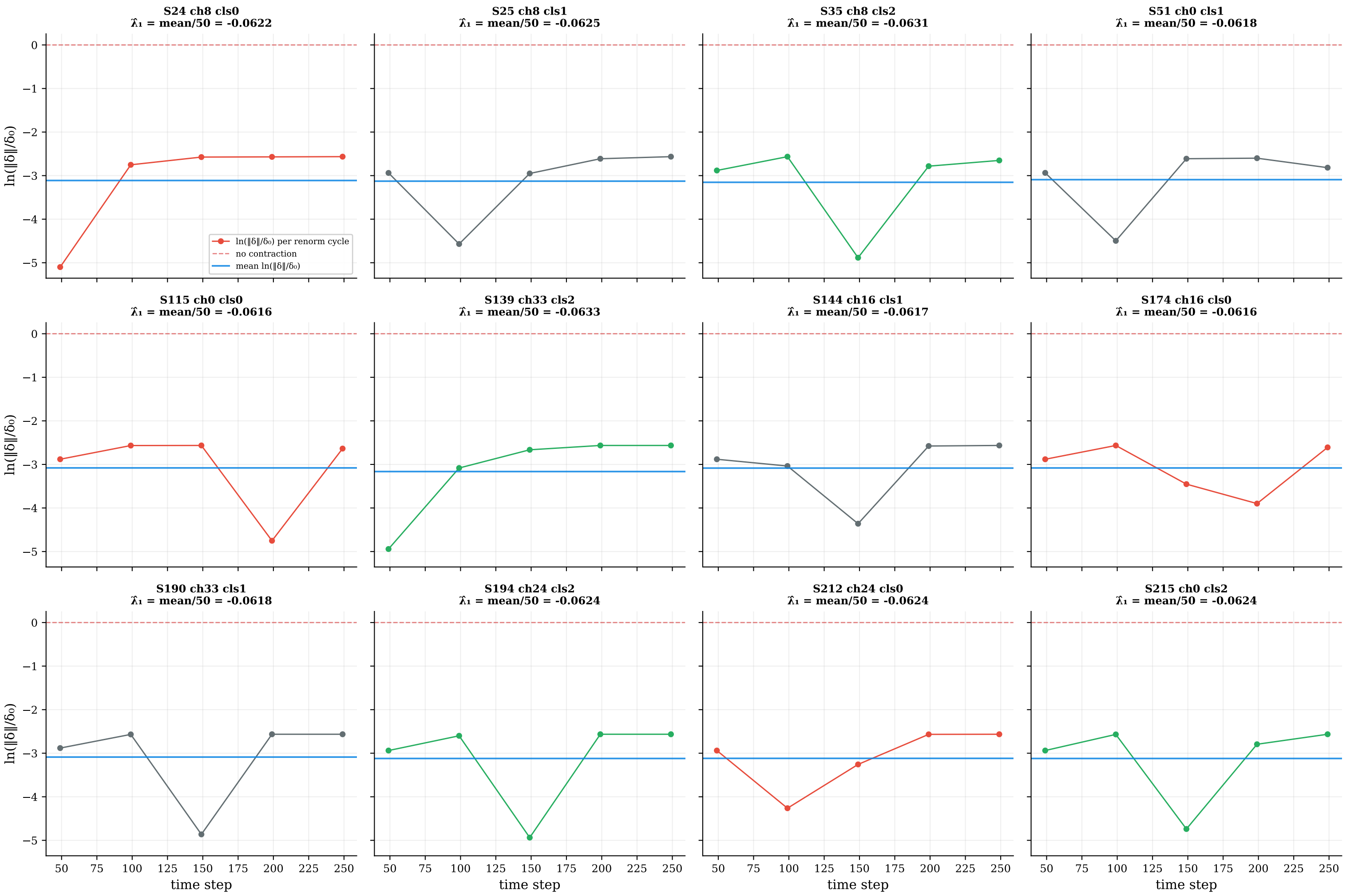
Raw Observation A.1a — Recurrent weight matrix W_{rec}



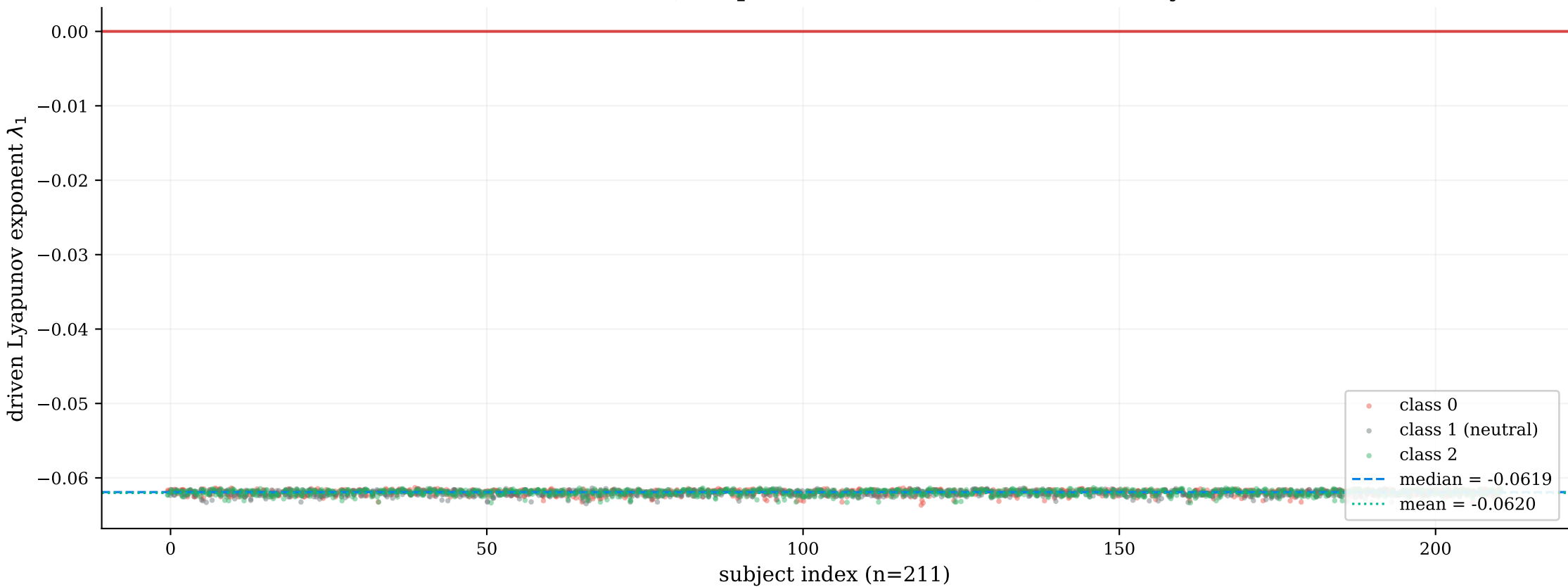
Raw Observation A.1b — Eigenspectrum of W_{rec} in the complex plane



Raw Observation A.1c — Sample Benettin trajectories (where each λ_1 comes from)

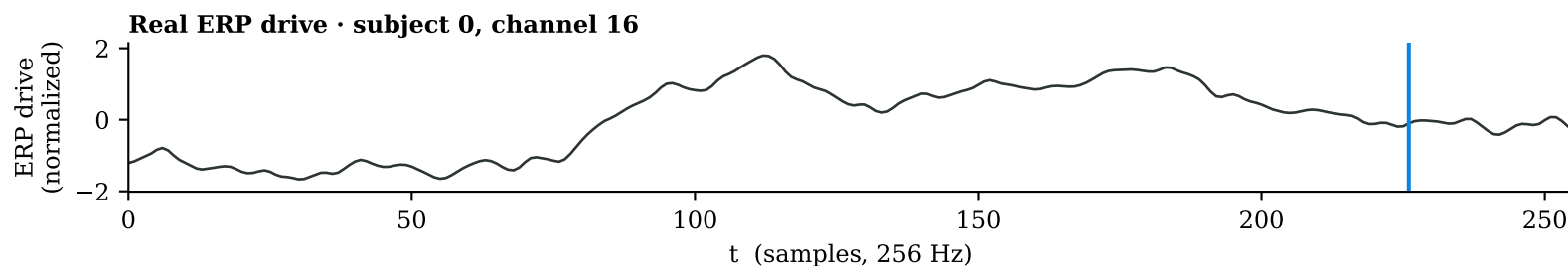
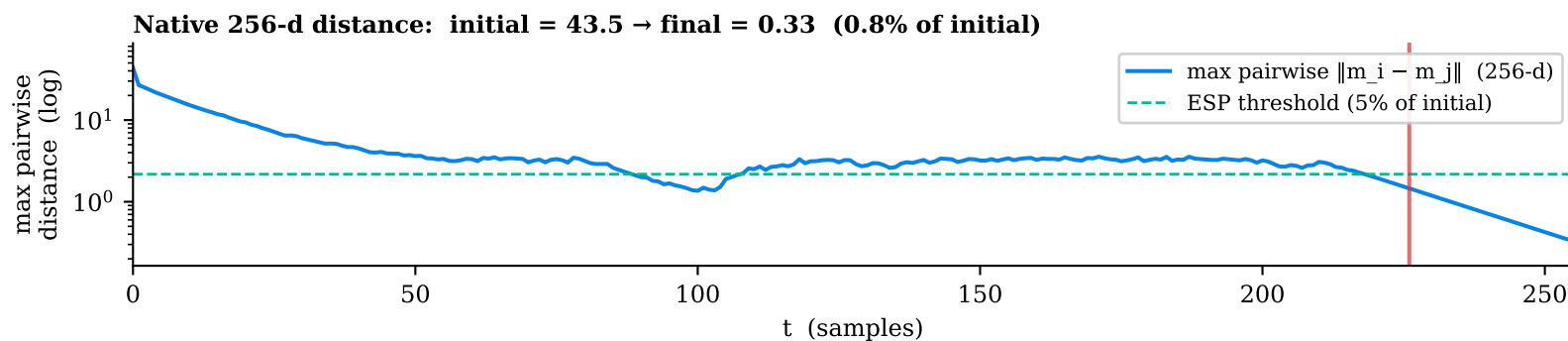
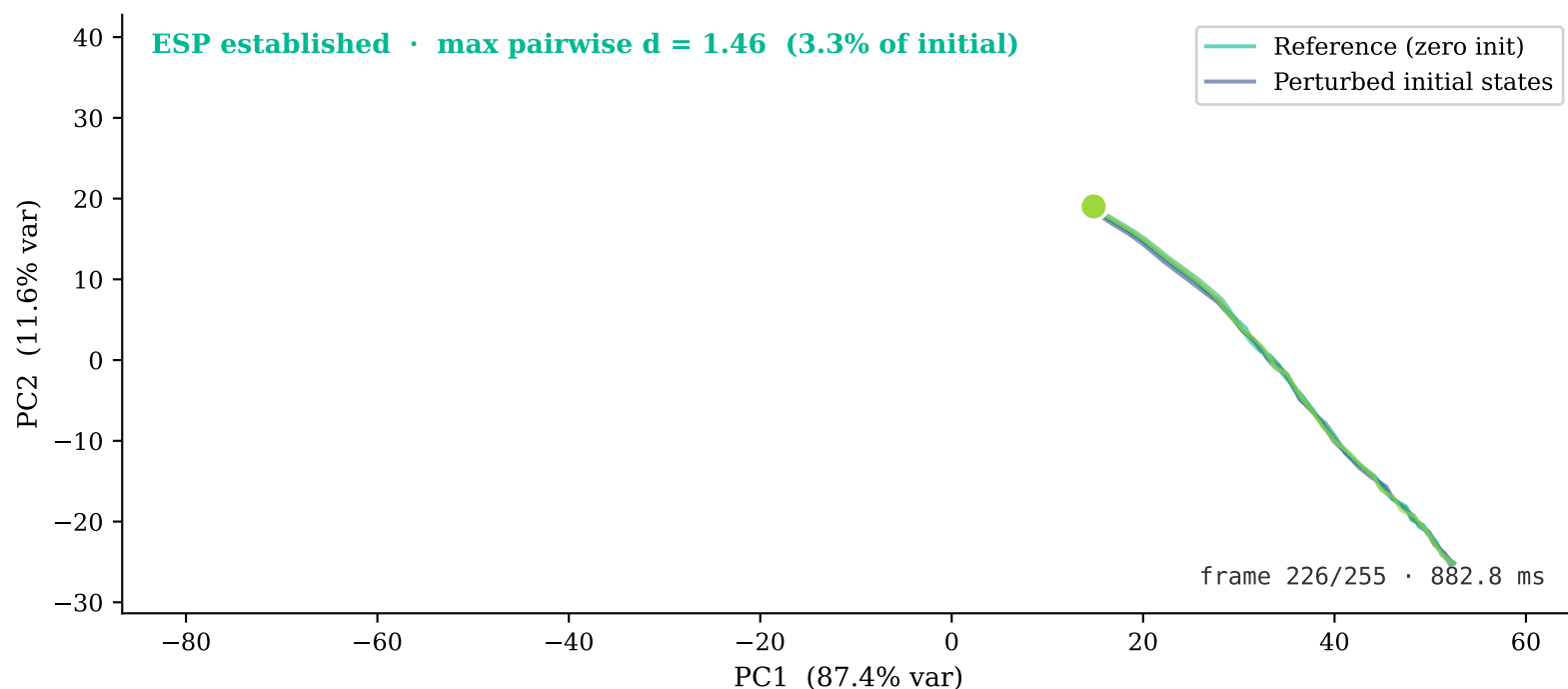


Raw Observation A.1d — All 3,165 per-trial measurements, colored by stimulus class



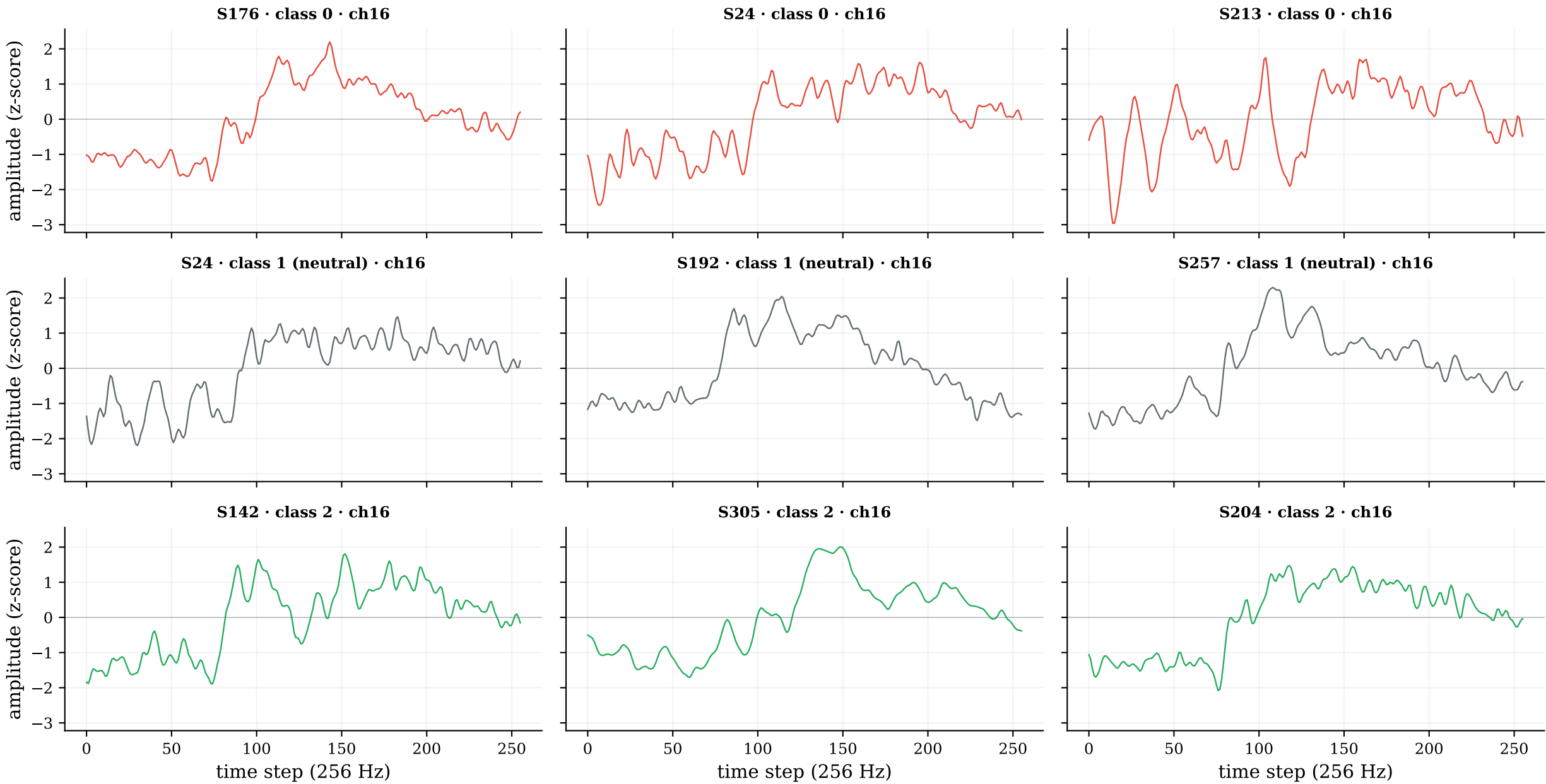
Reservoir Contraction — Echo State Property in motion

5 trajectories from different initial conditions, driven by the same real ERP.

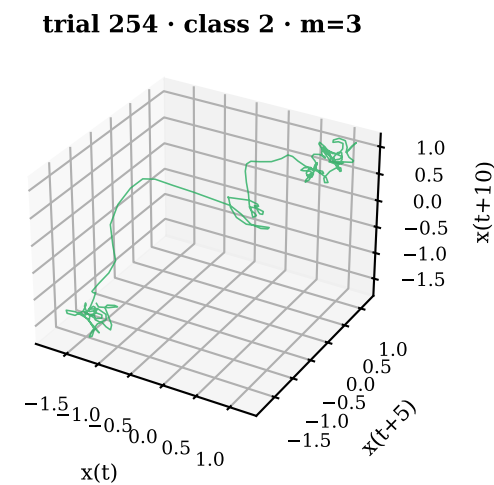
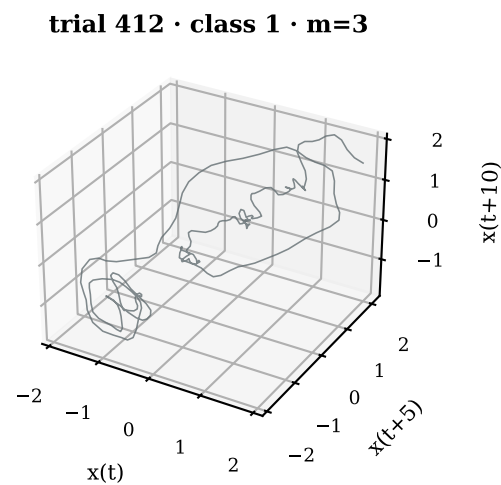
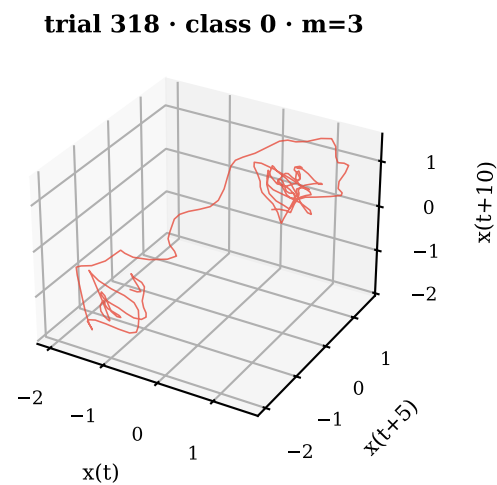
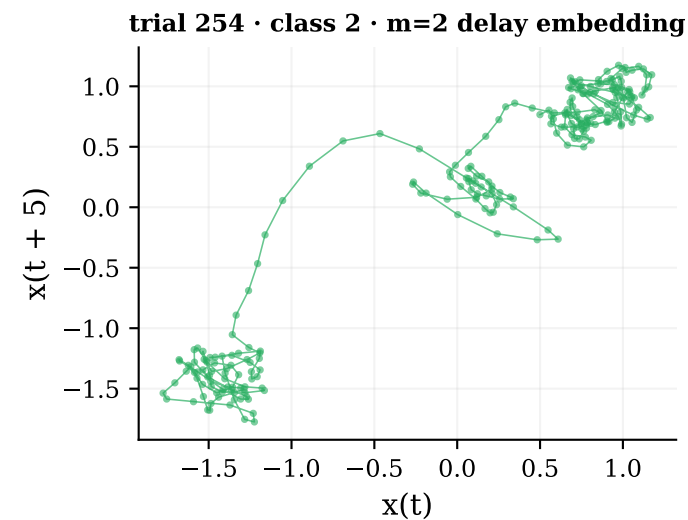
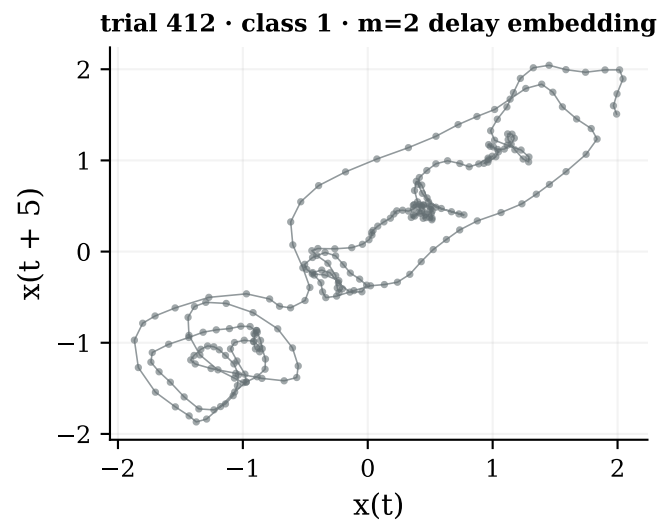
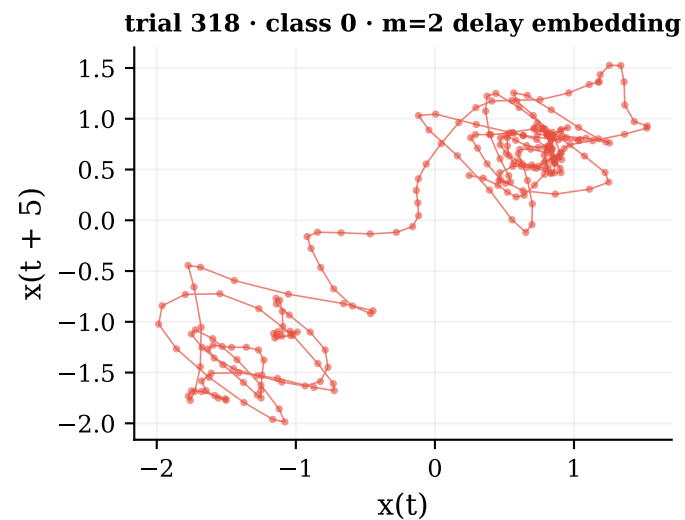


Driven contraction visualized in PCA-2D of the 256-d membrane state. All trajectories converge onto a single input-driven manifold (Echo State Property).

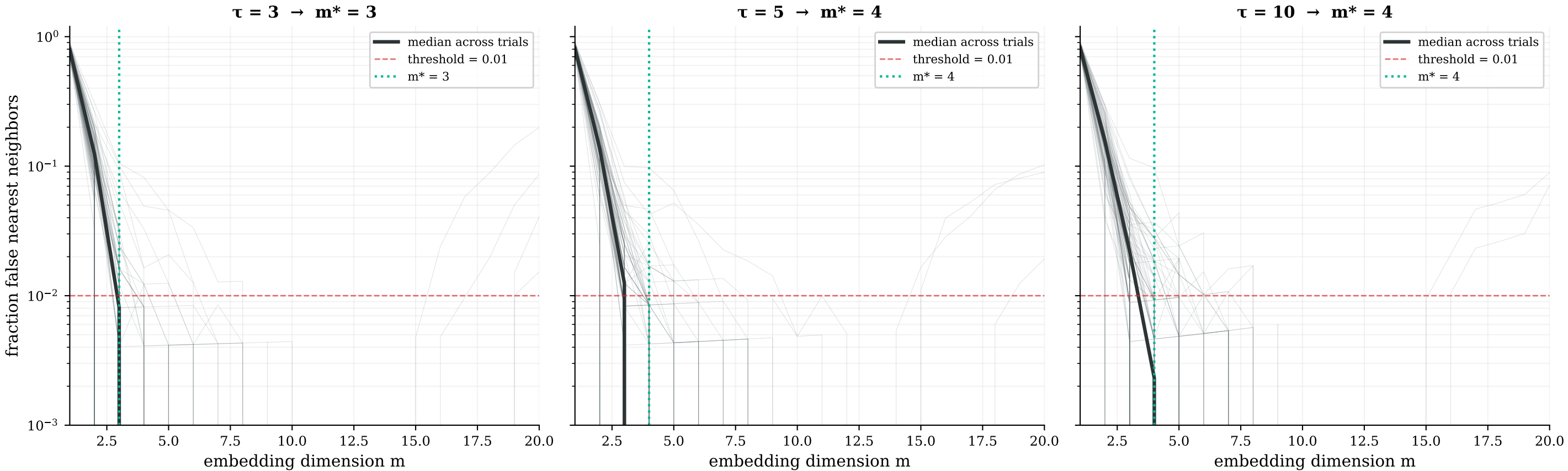
Raw Observation B.2a — Sample SHAPE ERPs (channel 16). These are the time series whose latent attractor we embed.



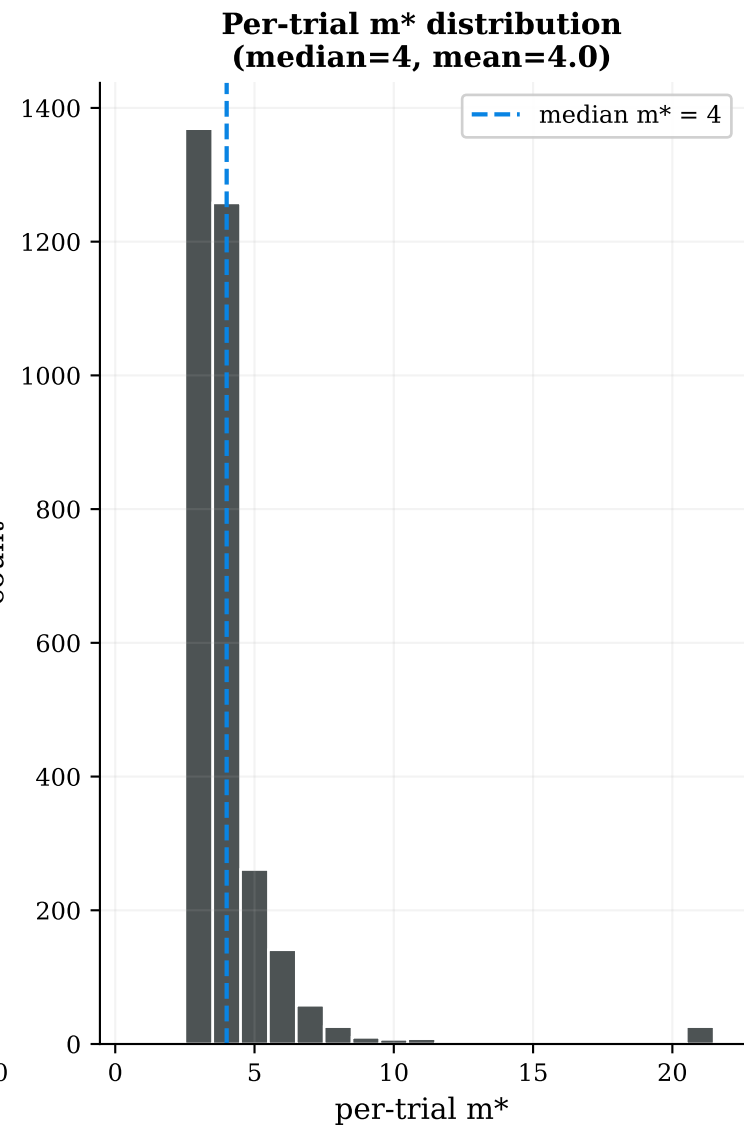
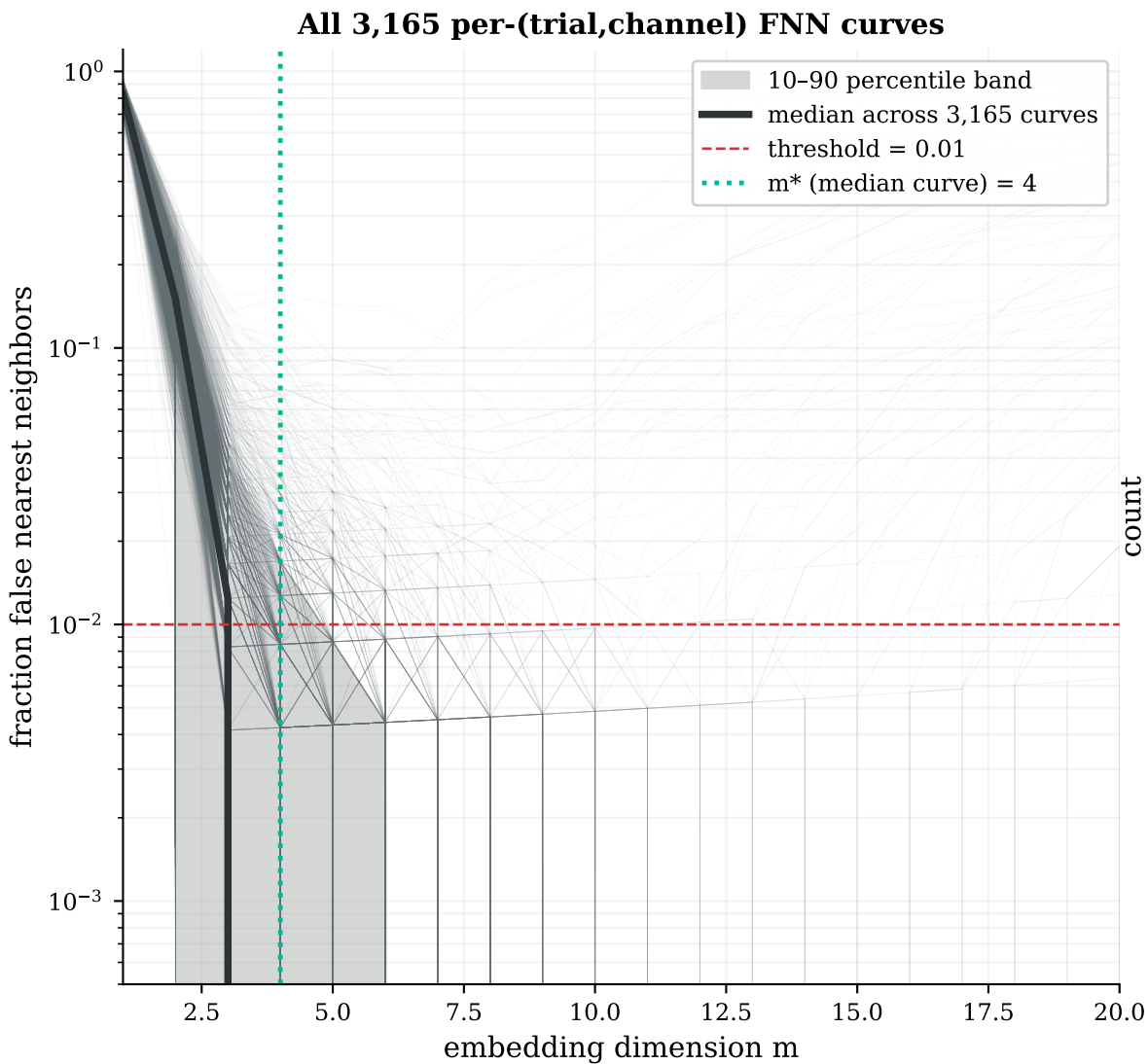
Raw Observation B.2b — Delay-embedded ERP trajectories ($\tau=5$). Top row: $m=2$. Bottom row: $m=3$. The geometry on which FNN operates.



Raw Observation B.2c — m^* robust across delay τ ($\tau=3 \rightarrow m^*=3$, $\tau=5 \rightarrow m^*=4$, $\tau=10 \rightarrow m^*=4$). The choice of τ does not rig the answer.

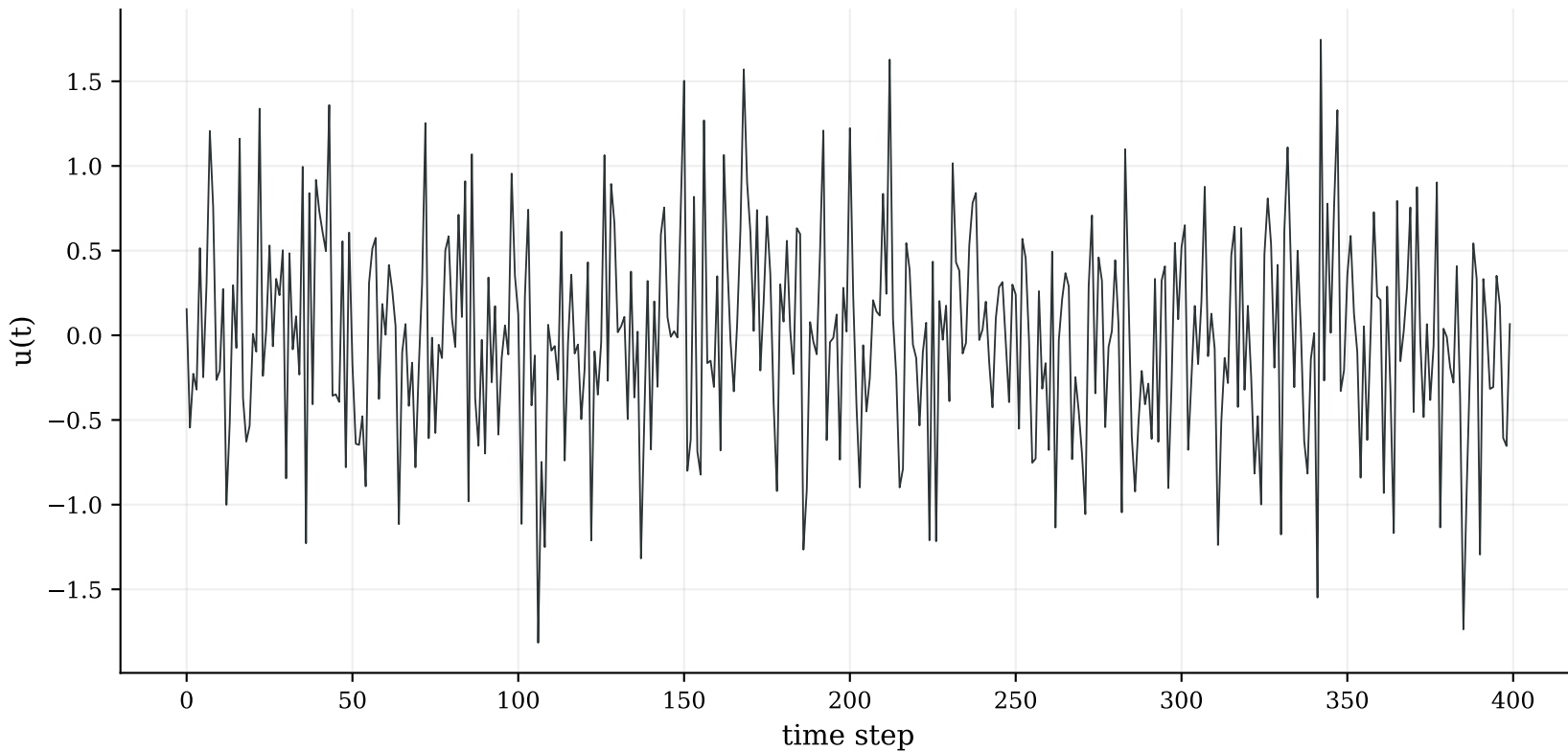


Raw Observation B.2d — Population of FNN curves: m^* is not a pooling artifact

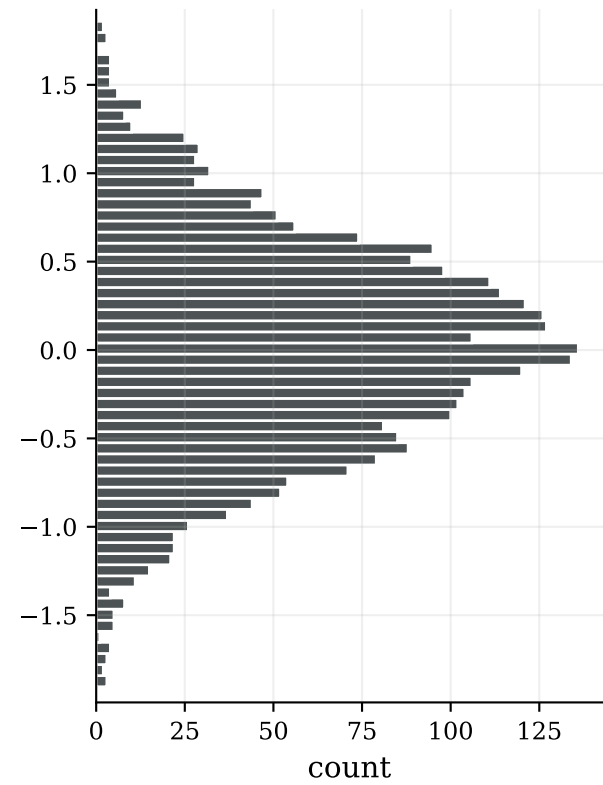


Raw Observation C.3a — Input drive for memory-capacity estimation

White-noise drive (first 400 steps shown)

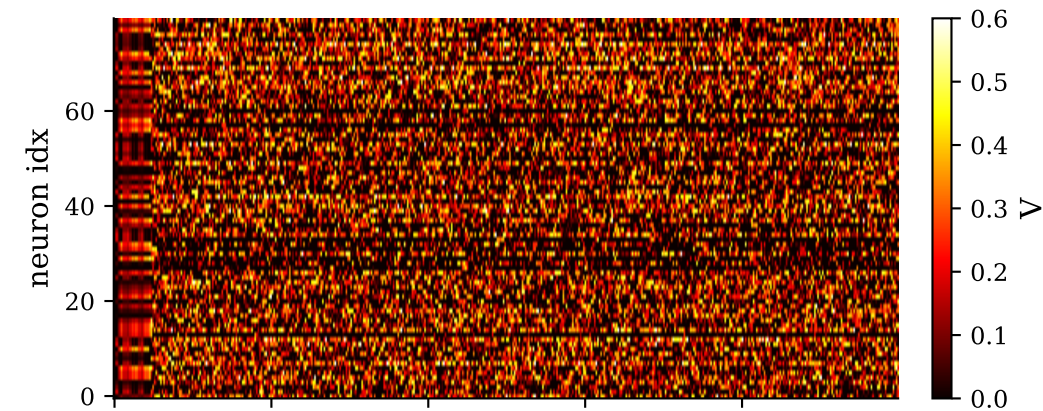
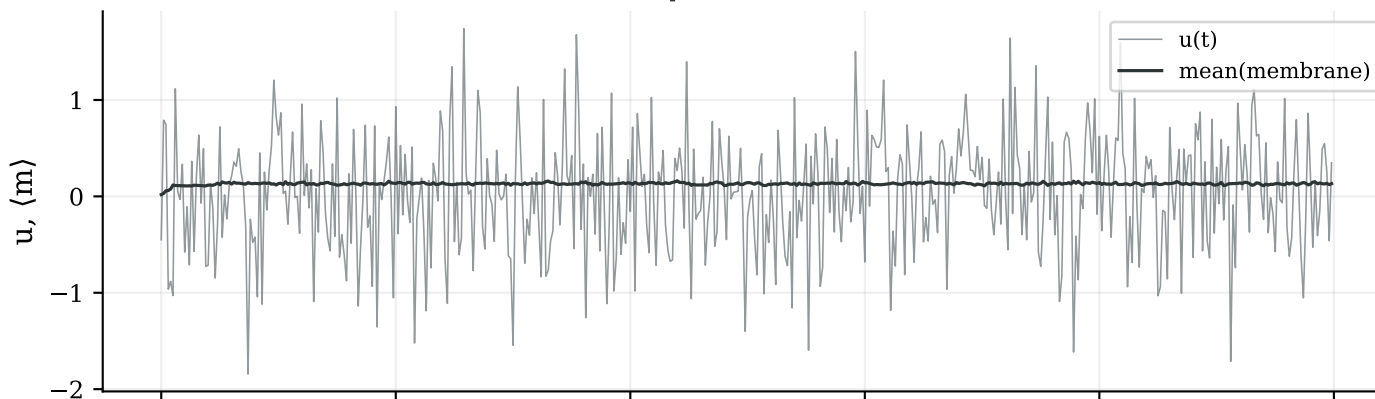


distribution
 $\mu=0.014$, $\sigma=0.598$

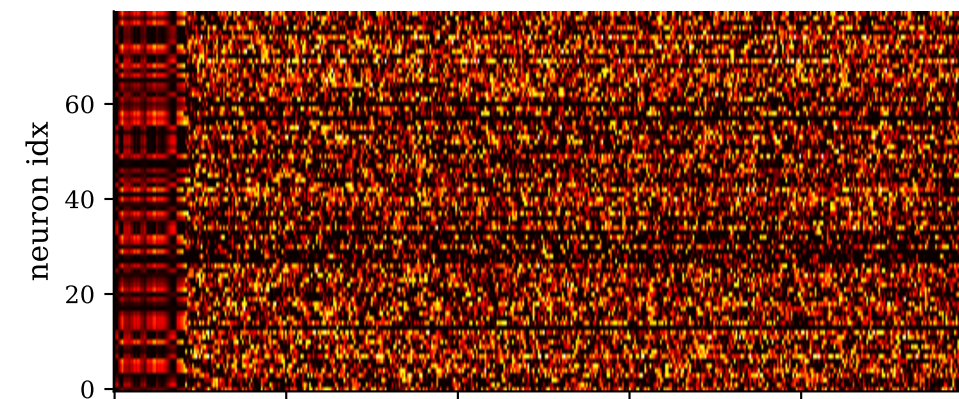
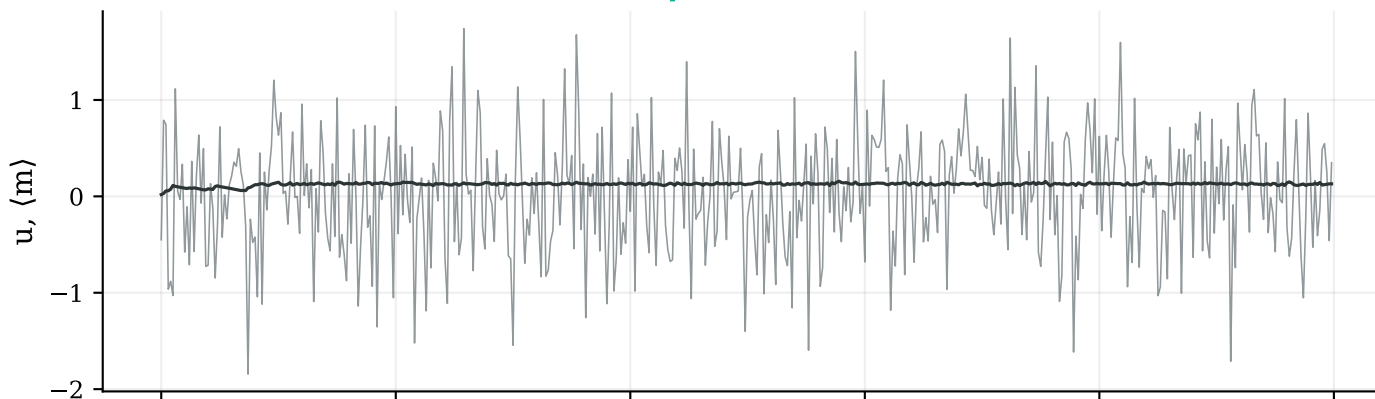


Raw Observation C.3b — Reservoir response at low / chosen / high β . $\beta = 0.05$ (chosen) shown in green.

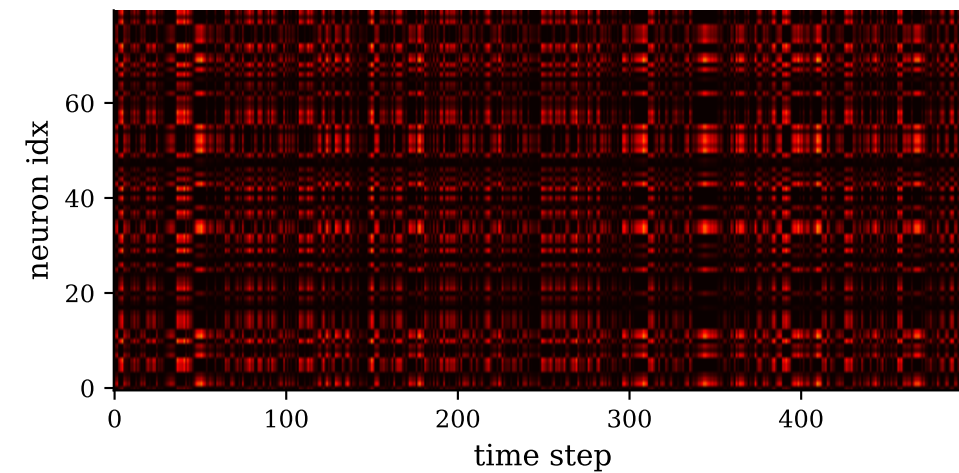
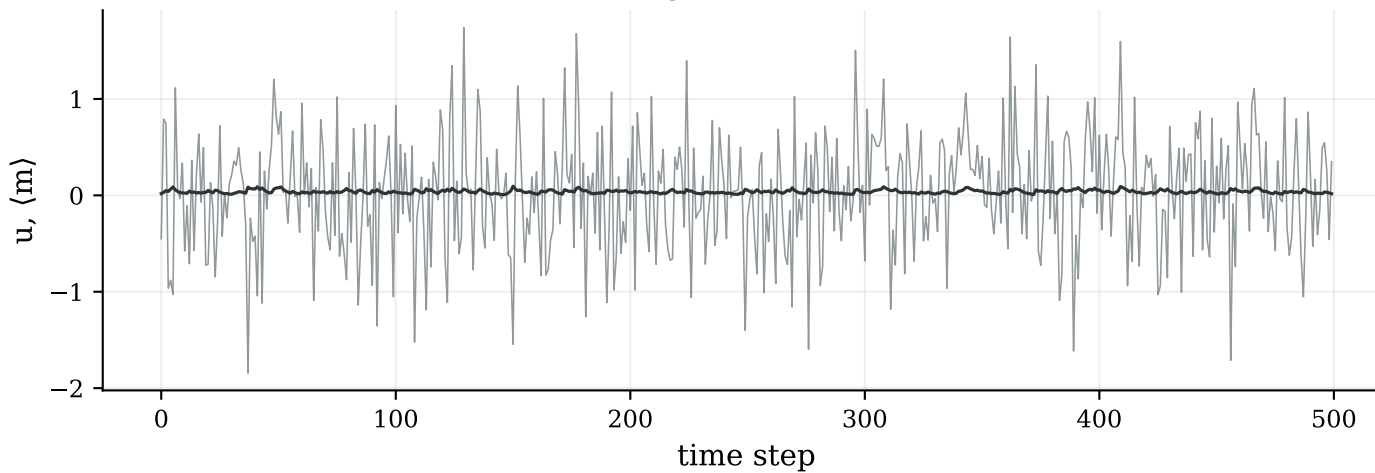
$\beta = 0.010$



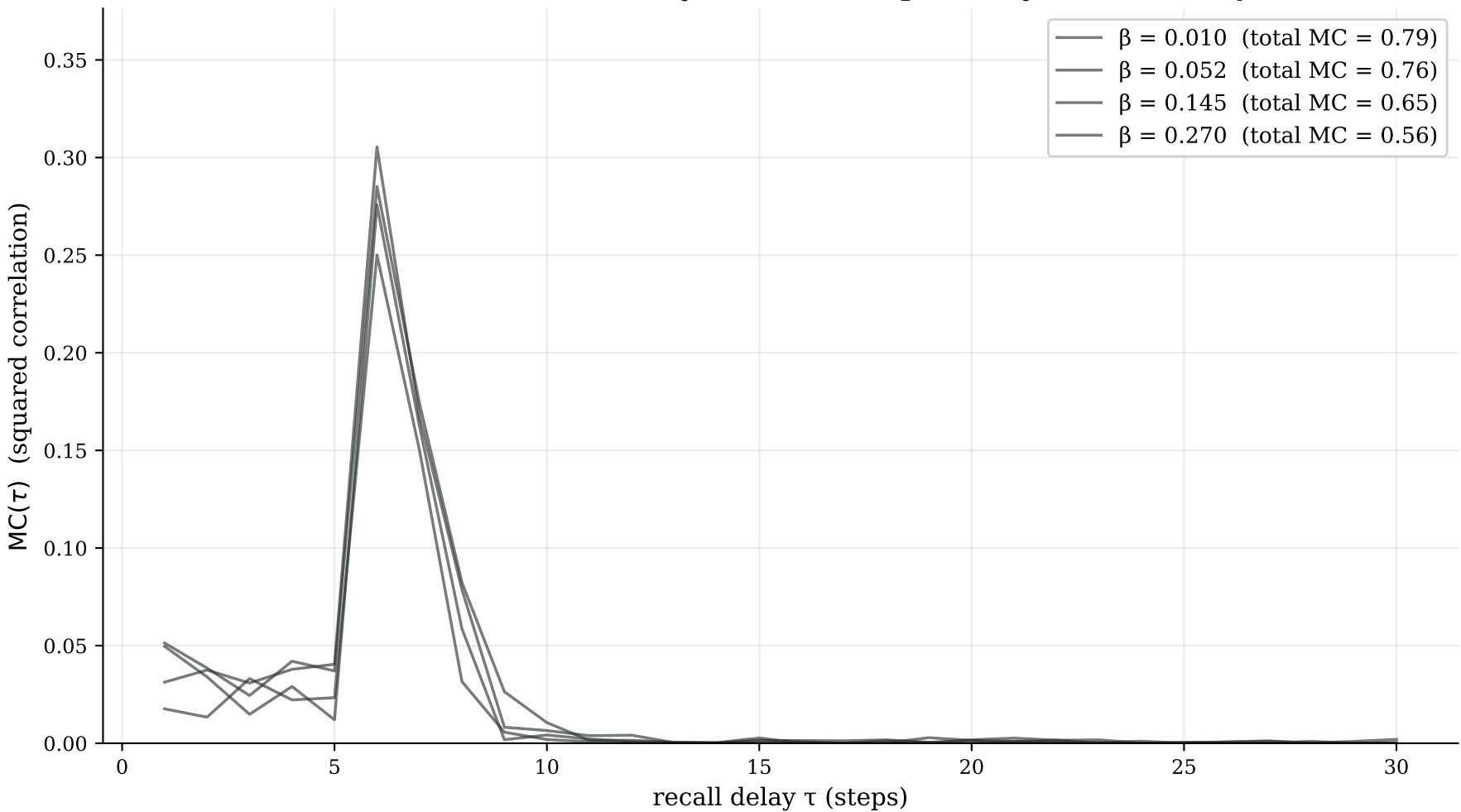
$\beta = 0.050$



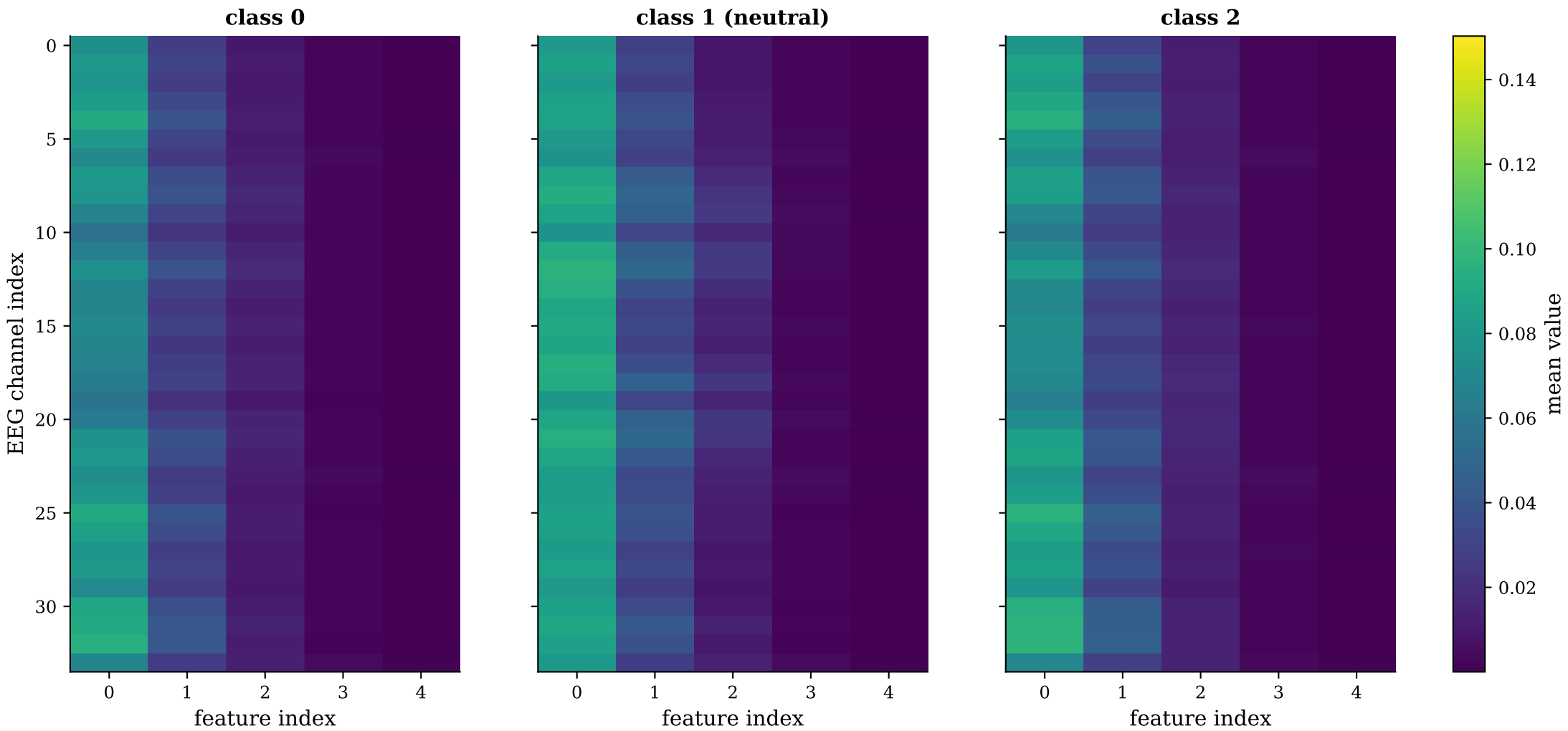
$\beta = 0.300$



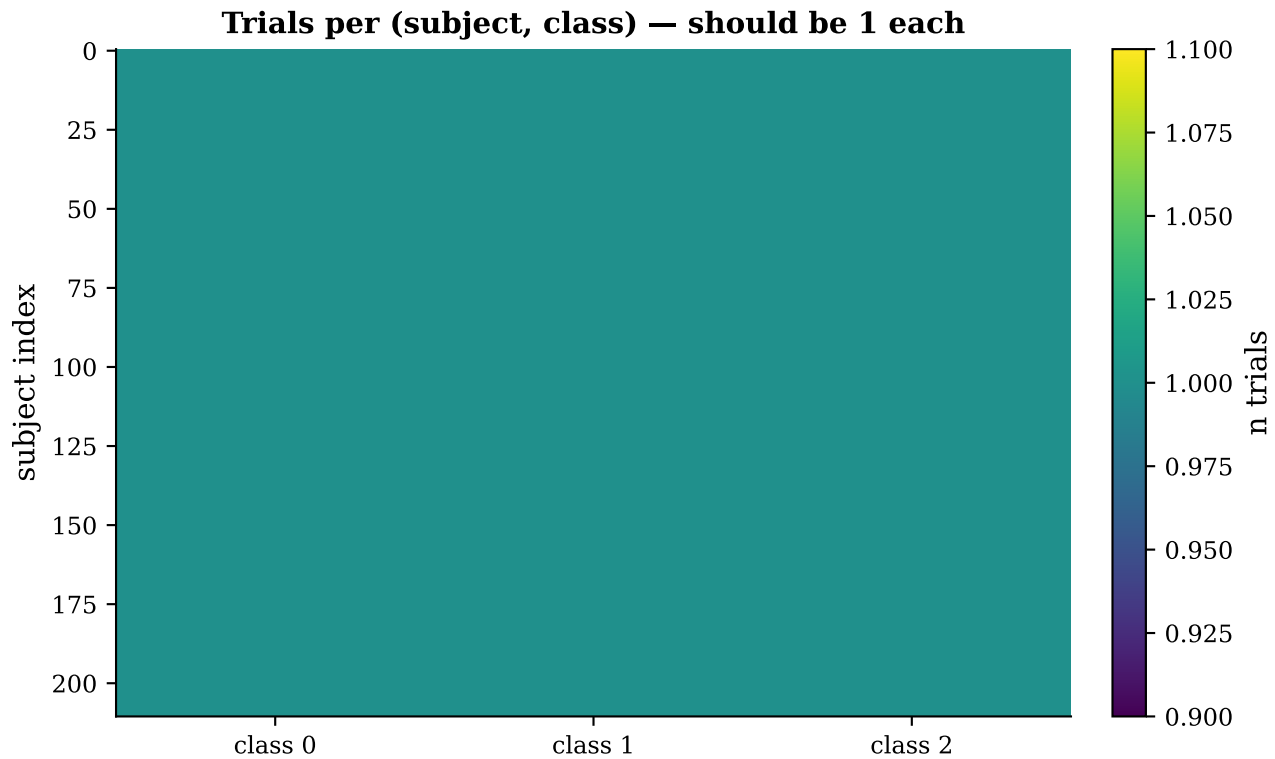
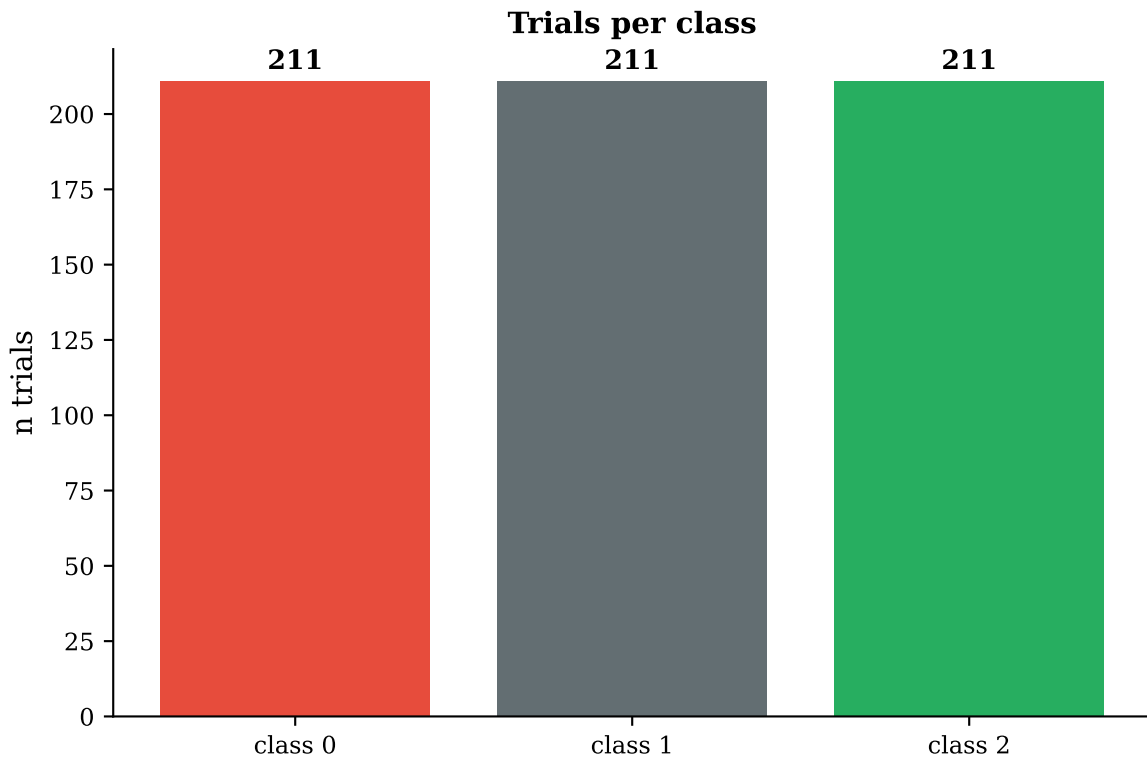
Raw Observation C.3c — Memory contribution per delay τ , at several β values



**Raw Observation D.4a — `conv\_feats`, per-class mean over trials
(this is the tensor whose channel axis the permutation null shuffles)**



Raw Observation D.4b — Class & subject distribution. n_subjects = 211, n_trials = 633.



Raw Observation D.4c — What channel permutation does: class 1 mean tensor (left) and two permutations (centre, right).
Within each column (feature), the channel ordering is reshuffled — same per-trial.

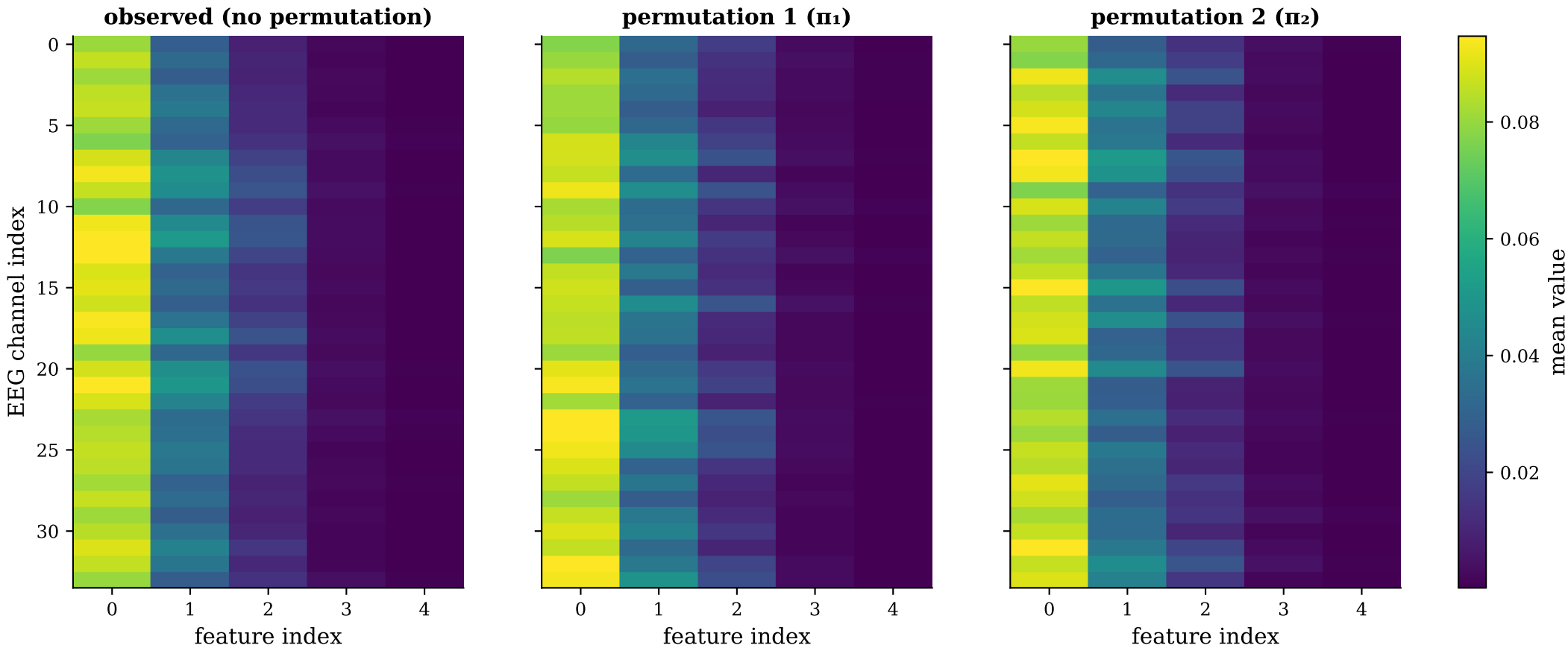


Figure-to-Source Index (Backup Appendix)

Committee due-diligence reference. Printed-only; not presented in the spoken deck.

Figure	Script / output path	Chapter	Notes
K	figure_K_opening_questions/	Ch. 1	Opening questions
IP	figure_IP_intellectual_provenance/	Ch. 2	Scholarly-lineage map
F	figure_F_theorem_scaffold/	Ch. 3	Theorem scaffold (Takens / Jaeger / Maass / Kennel-Brown)
MR	figure_MR_methodological_refusals/	Ch. 5–6 (transversal)	Methodological refusals — rides on Exp D / C anchors
Exp A	experiment_a_autonomous_vs_driven/	Ch. 6	Autonomous $p(W) \rightarrow$ driven Lyapunov λ_1 measured on 3,165 trials
Exp B	experiment_b_takens_dimension/	Ch. 4 / Ch. 6	Takens-motivated question, FNN-measured embedding dimension
Exp C	experiment_c_memory_capacity/	Ch. 6	Propagation Operating Characteristic; $\beta = 0.05$ vs measured β^*
TB	figure_TB_theoretical_bounds/	Ch. 6 (reads A / C)	Analytic bounds vs measured operating points
Exp D	experiment_d_channel_permutation/	Ch. 5	Per-trial channel-permutation null; stimulus-class demonstration
G	figure_G_failure_gallery/	Ch. 5–6 + audit	Failure-to-pivot narrative across A, C, D
J	figure_J_contributions/	Ch. 7	Five named contributions
AM	figure_AM_anchor_map/	—	This slide (self-reference; no anchor)
OQ	figure_OQ_open_questions/	Ch. 8	Open questions / closing
AC	figure_AC_research_provenance/	—	Research-provenance disclosure (transversal)

Section and page anchors are intentionally omitted: they are brittle across LaTeX edits. Author fills in finer granularity from the stable compiled dissertation PDF as required.